

Case Study Analysis Report

Assignment Title: Intelligent AI-Based Smart Aquaculture Disease Prediction and Feed Optimization System

CO1-ASSESSMENT TOOL - 1

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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions and constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Marks
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/25

1. Scenario Understanding and Problem Analysis

1.1 Industry Context

Aquaculture is one of the fastest-growing sectors in global food production. Fish farming plays a vital role in meeting the increasing demand for seafood. However, maintaining fish health, monitoring water quality, and ensuring efficient feed management are major challenges. Modern technologies such as Artificial Intelligence (AI), Internet of Things (IoT), underwater sensors, and computer vision help automate monitoring, improve productivity, reduce losses, and support sustainable aquaculture practices.



1.2 Problem Statement

Fish farms often experience production losses due to disease outbreaks, poor water quality, and inefficient feeding practices. Manual monitoring is time-consuming and cannot continuously observe fish health or environmental changes. An AI-powered smart aquaculture system is required to continuously monitor water quality, analyze fish behaviour using computer vision, predict disease outbreaks, optimize feeding schedules, and generate farm management reports to improve efficiency and reduce operational costs.

1.3 Stakeholders

- Fish Farm Owner
- Farm Manager
- Farm Workers
- Veterinarian
- System Administrator
- IoT Sensor Maintenance Team
- AI System Developer

1.4 Existing Challenges

- Late detection of fish diseases
- Poor water quality monitoring
- High feed wastage
- Manual farm management
- Lack of real-time monitoring

- Increased operational costs
- Low production efficiency
- Inaccurate disease diagnosis

1.5 System Objectives

- Monitor fish health continuously.
- Predict disease outbreaks using AI.
- Analyze water quality parameters.
- Optimize feeding schedules.
- Detect abnormal fish behavior.
- Reduce feed wastage.
- Generate farm performance reports.
- Improve aquaculture productivity.



2. Functional Requirements

- The system shall collect water quality data from IoT sensors.
- The system shall monitor fish behavior using underwater cameras.
- The system shall predict fish diseases using AI algorithms.
- The system shall recommend optimized feeding schedules.
- The system shall generate alerts for abnormal conditions.
- The system shall generate daily, weekly, and monthly reports.
- The system shall maintain fish health records.
- The system shall allow user login and authentication.
- The system shall display real-time farm status.
- The system shall store historical sensor data.



3. Non-Functional Requirements

Performance

- Real-time monitoring with fast response time.

Reliability

- Continuous operation with minimal downtime.

Security

- Secure login and encrypted data storage.

Availability

- System available 24×7.

Scalability

- Support multiple ponds and fish farms.

Usability

- User-friendly dashboard for farmers.

Maintainability

- Easy software maintenance and updates.

4. Actor Identification

Primary Actors

- Fish Farm Owner
- Farm Manager
- Farm Worker
- Veterinarian
- Administrator

Secondary Actors

- AI Prediction Engine
- IoT Water Sensors
- Underwater Camera
- Notification Service

5. Use Cases

- Login
- Monitor Water Quality
- Monitor Fish Health
- Predict Disease
- Analyze Fish Behaviour
- Optimize Feed Schedule
- Receive Alerts
- Generate Reports
- Manage Users
- Logout



6. UML Use Case Diagram

(Insert the UML Use Case Diagram here.)

Actors:

- Farm Owner
- Farm Manager
- Farm Worker
- Veterinarian
- Administrator

Use Cases:

- Login
- Monitor Water Quality
- Predict Disease
- Optimize Feeding
- View Dashboard
- Generate Reports
- Manage Users

7. Use Case Descriptions

Use Case: Login

Actor: Farm Owner/Admin

Precondition: User must be registered.

Flow:

1. Enter username.
2. Enter password.
3. System verifies credentials.
4. Dashboard is displayed.

Postcondition: User successfully logs in.

Use Case: Predict Disease

Actor: AI Prediction Engine

Flow:

1. Read sensor data.
2. Analyze fish behavior.
3. Predict disease.
4. Send notification.

Use Case: Optimize Feed Schedule

Actor: Farm Manager

Flow:

1. Collect feeding history.
2. AI analyzes data.
3. Recommend feeding schedule.
4. Save recommendation.

8. Software Requirement Specification (SRS)

8.1 Problem Description

The proposed AI-based smart aquaculture system continuously monitors fish health and water quality using IoT sensors and computer vision. It predicts disease outbreaks, recommends feeding schedules, and generates reports to improve farm productivity and reduce operational costs.

8.2 Scope of the System

The system supports:

- Fish health monitoring
- Water quality monitoring
- Disease prediction
- Feed optimization
- Report generation
- Alert notifications
- Farm management dashboard



8.3 Data Requirements

The system stores:

- Fish ID
- Pond ID
- Water Temperature
- pH Level
- Dissolved Oxygen
- Ammonia Level
- Feeding Schedule
- Disease History
- User Information
- Sensor Readings
- Reports

8.4 External Interface Requirements

Hardware Interface

- IoT Sensors
- Underwater Camera
- Server
- Computer
- Mobile Device

Software Interface

- AI Prediction Model
- Database Management System
- Web Application
- Mobile Application

Communication Interface

- Wi-Fi
- Internet
- Cloud Storage

8.5 System Features

- AI-based Disease Prediction
- Water Quality Monitoring
- Fish Behaviour Analysis
- Smart Feed Recommendation
- Alert Notification System
- Report Generation
- Dashboard Monitoring
- Historical Data Analysis

9. Assumptions

- Internet connectivity is available.
- Sensors function properly.
- Cameras continuously monitor fish.
- Users have basic computer knowledge.
- AI model is trained using accurate data.

10. Constraints

Technical Constraints

- Sensor failures may affect prediction accuracy.

Operational Constraints

- Requires trained farm personnel.

Economic Constraints

- High installation and maintenance costs.

Legal Constraints

- Must comply with environmental and aquaculture regulations.

Time Constraints

- Sensor data should update every few seconds.

11. Requirement Validation

- All stakeholder requirements have been identified.
- Functional requirements are complete and consistent.
- Non-functional requirements satisfy system quality.
- Actors and use cases are clearly defined.
- Requirements are clear, complete, and unambiguous.
- System scope meets project objectives.

12. Conclusion

The **Intelligent AI-Based Smart Aquaculture Disease Prediction and Feed Optimization System** provides an efficient solution for modern fish farming by integrating Artificial Intelligence, IoT sensors, and computer vision technologies. The system improves fish health monitoring, predicts disease outbreaks at an early stage, optimizes feeding schedules, reduces operational costs, and enhances overall farm productivity. It supports sustainable aquaculture practices through real-time monitoring and data-driven decision-making.

13. References

1. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson.
2. Roger S. Pressman and Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*, McGraw-Hill.
3. FAO (Food and Agriculture Organization). *The State of World Fisheries and Aquaculture*. <https://www.fao.org>
4. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.